

Rock volume estimation from incomplete point clouds using interpretable machine learning for rockfill material gradation analysis

Jiayu Wang^{a,1}, Mengnan Shi^{a,1}, Yilong Wu^b, Peng Ji^a, Hongtao Li^a, Qiang Yao^{a,c,*}

^a State Key Laboratory of Hydraulics and Mountain River Engineering, College of Water Resource and Hydropower, Sichuan University, Chengdu, Sichuan 610065, China

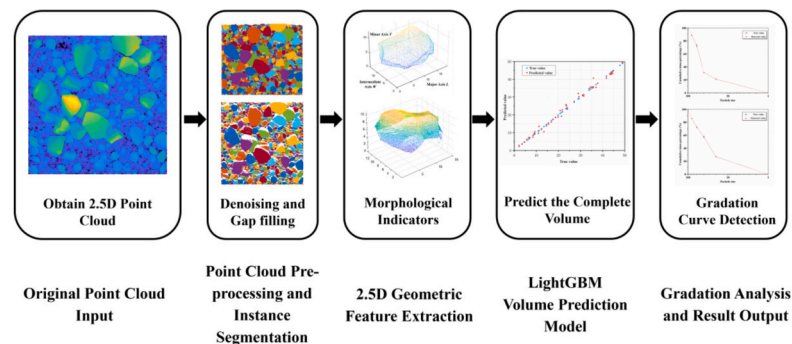
^b Sinohydro Bureau 7 Co., Ltd, Chengdu, Sichuan 610213, China

^c Technology Innovation Center for Risk Prevention and Mitigation of Geohazard, Ministry of Natural Resources, Chengdu, Sichuan 611734, China

HIGHLIGHTS

- Proposed a rock volume prediction method using 2.5D point clouds and interpretable machine learning.
- Achieved high accuracy with $R^2 = 0.979$ and $MAPE = 9.39\%$ using a LightGBM model.
- SHAP analysis revealed particle volume and surface area as key predictive features.
- Automated gradation detection workflow showed less than 2% error across all size ranges.

GRAPHICAL ABSTRACT



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ABSTRACT

Accurate particle size distribution and volume estimation are fundamental to understanding the behavior of granular materials like rockfill in geotechnical applications. However, conventional sieving is labor-intensive, and 2D imaging methods fail to capture true 3D morphology, introducing systematic bias in volume prediction. While 3D laser scanning offers richer geometric data, real-world scans yield only partial (“2.5D”) point clouds due to occlusion, severely limiting direct volume calculation. To address this, we propose an interpretable machine learning framework for estimating complete particle volumes from incomplete 2.5D point clouds, enabling high-accuracy gradation analysis. A dataset of 400 paired 2.5D and complete 3D particles was constructed from real rockfill materials. Morphological analysis revealed consistent shape distributions across sizes, supporting a generalized model. Using LightGBM, we developed a volume prediction model solely from 2.5D-derived 3D features, achieving exceptional accuracy ($R^2 = 0.979$, $MAPE = 9.39\%$). SHAP analysis provided physical interpretability, identifying particle volume and surface area as dominant predictors. Integrated into a full pipeline — including segmentation and size classification — the automated workflow was validated on six synthetic stockpile cases generated from realistic morphological distributions, yielding gradation errors less than 2% across all size fractions, with predicted curves closely matching ground truth.

* Corresponding author at: State Key Laboratory of Hydraulics and Mountain River Engineering, College of Water Resource and Hydropower, Sichuan University, Chengdu, Sichuan 610065, China.

E-mail address: yaoqiang777@126.com (Q. Yao).

¹ Jiayu Wang and Mengnan Shi contributed equally to this work.

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1. Introduction

In large-scale geotechnical engineering projects such as rockfill dams, gradation control of fill materials constitutes a critical factor ensuring structural safety and durability [1–5]. However, conventional gradation detection primarily relies on manual sieving and weighing—a process that is labor-intensive, inefficient, and inadequate for meeting modern engineering demands for high-frequency, real-time monitoring, and construction intelligence [6]. Therefore, it is of great significance to realize the automatic analysis of the gradation of rockfill materials. Accurate gradation data not only ensures compliance with design specifications but also directly influences the hydraulic conductivity, compressibility, and interface behavior of rockfill layers—key factors affecting the service life and safety of infrastructure.

Currently, some automatic gradation detection methods use image recognition technology. By extracting the major and minor axis dimensions of particles in the image, the particles are simplified into an ellipsoid model, and the volume is estimated accordingly [7–10]. For example: Fan, et al. [11] proposed a gradation detection method for rockfill materials based on the improved Mask R-CNN, which integrates ResNeXt101 and the squeeze-excitation module. The average precision reaches 0.934, the intersection-over-union ratio is 0.879, and on-site gradation analysis can be completed within 60 s, effectively supporting the construction of earth-rock dams; Fan, et al. [12] proposed a gradation detection method for rock materials based on UAV images, constructed the Rock-net model to achieve multi-scale rock material segmentation, and further constructing a regression prediction model using XGBoost optimized by the Crested Porcupine Optimizer (CPO), referred to as CPO-XGB, to capture the nonlinear relationship between morphological features and spatial gradation. The R^2 is 0.977, and the FPS is 12.8 FPS. It has good generalization ability and is suitable for the detection of different particle size gradations; Zhou, et al. [13] proposed a gradation detection method for rock materials based on video image recognition. Through the video acquisition device and recognition algorithm, dynamic and continuous detection is realized, and the accuracy reaches 93%; Yao, et al. [14] proposed a gradation detection method for rock materials based on infrared images, and combined image enhancement and the improved watershed algorithm to achieve particle segmentation and recognition. The actual measurement shows that the average errors of the gradation detection of pebbles and crushed stones are 4% and 5% respectively; Pang, et al. [15] proposed a method for identifying the particle size distribution of rock cuttings based on CNN, constructed an image data set and optimized the model structure, and finally the R^2 reached 0.923.

Although the above-mentioned image recognition methods have made significant progress in segmentation accuracy and efficiency, since they are essentially two-dimensional projections, it is difficult to capture the height information of particles, and it is impossible to truly restore the three-dimensional shape of complex and irregular particles, resulting in systematic deviations in volume estimation, which in turn affects the reliability of gradation analysis [16,17]. In contrast, 3D point cloud technology can capture more abundant spatial geometric information, providing new possibilities for improving the accuracy of volume estimation [18–22]. Existing studies have obtained complete particle point clouds through individual scanning under experimental conditions and calculated their volumes for gradation analysis. For example: Faramarzi, H and Esmaeili [23] proposed a method for identifying the particle size of crushed rocks based on laser point clouds. They used synthetic data to train a deep model to achieve 3D particle segmentation, and the average recognition accuracy in three groups of measured point clouds exceeded 92%. This method does not rely on RGB information and is suitable for complex lighting environments; Mineo, et al. [24] proposed a method for estimating the volume of rock blocks based on TLS point clouds, used CloudCompare to semi-automatically process the data, and the volume estimation results of unstable rock blocks conformed to the lognormal distribution, with an estimation accuracy of 95%; Yang, et al. [25]

proposed an automatic detection method for the gradation of coarse aggregates based on 3D point clouds, combined with the ISRNet model to achieve particle size classification. The accuracies within the particle size ranges of 0–10 mm and 10–20 mm are 99.12% and 98.89% respectively, and the maximum error of gradation evaluation is 1.63%. Ding et al. [26] proposed an equivalent particle size and established a predictive framework by integrating 3D scanning with supervised learning, enabling accurate characterization and prediction of coarse aggregate morphological and gradation features. While recent learning-based methods have attempted to estimate volumes from 2D or partial 3D data, they often treat volume prediction as a black-box regression task. These approaches rely heavily on large-scale training datasets without offering physical interpretability, and many adopt simplified shape assumptions or require multi-view imaging setups impractical in real construction contexts. It must be emphasized that most existing studies operate on complete point clouds obtained in laboratory environments—a condition starkly different from actual stockpile scenarios where pervasive inter-particle occlusion occurs. In such field conditions, laser scanning typically captures only upper-surface point clouds, resulting in “2.5D” incomplete data. Although the 2.5D point cloud is incomplete due to occlusion, it still retains the real three-dimensional spatial coordinate information, which is essentially different from the two-dimensional contour expression based on perspective projection in traditional image recognition.

Recognizing this unique characteristic, several studies have begun exploring the potential of 2.5D representations for 3D geometric inference. Hamid Ashfaq et al. [27] proposed a machine learning approach utilizing 2.5D features—such as silhouette, depth, and surface normals—to enable efficient 3D shape reconstruction from a single RGB image, demonstrating how limited geometric cues can be leveraged for volumetric estimation. Similarly, Spreitzer et al. [28] further advanced this direction by employing SfM photogrammetry to generate 2.5D volume models, revealing systematic overestimation errors arising from convex hull approximation; notably, they innovatively proposed using the discrepancy between 2.5D and 3D volumes as a proxy for estimating porosity in granular media. Meanwhile, Yao et al. [29] adopted 2.5D sketches composed of depth and surface normal maps as an intermediate representation to bridge 2D images and 3D point clouds, enabling single-image 3D reconstruction through learned latent space mapping—an approach that explicitly addresses ambiguity in occluded regions. These efforts collectively highlight the growing recognition that meaningful 3D information can be inferred from incomplete 2.5D observations, provided appropriate modeling strategies are employed. Yet, despite these advances, there remains a critical gap in methods specifically designed to predict complete particle volumes directly from 2.5D point clouds for accurate gradation analysis in practical engineering settings.

Therefore, this paper proposes a particle volume prediction method based on incomplete point cloud features and explainable machine learning, establishing a comprehensive technical workflow for automated gradation detection of rockfill materials.

The main contributions are as follows:

- (1) Construction of a paired dataset comprising 400 sets of 2.5D and complete 3D particles, with extracted multiscale geometric features to support model training and validation;
- (2) Proposal of a LightGBM-based volume prediction framework leveraging 3D geometric information from 2.5D point clouds, significantly enhancing accuracy under occluded conditions;
- (3) Implementation of SHAP interpretability analysis to identify key predictive features, improving model transparency and applicability;
- (4) Development of a field-deployable automated gradation detection workflow with experimental verification confirming detection errors below 2% across all particle size ranges.

2. Methodology

In research on gradation detection of rockfill materials based on 3D point clouds, point cloud segmentation methods extract rock particle point clouds from stockpile scans to calculate particle size and volume. The volumetric proportion of particles within different size ranges is then statistically determined to achieve gradation calculation and analysis. However, current 3D laser scanning technology can only capture the upper morphology of particles ("2.5D particles"), failing to penetrate and capture lower surfaces. Consequently, the segmented rock particles retain only partial surface point clouds, leading to significant errors when calculating volumes using mainstream convex hull methods, thereby increasing gradation detection inaccuracies.

To address this limitation, this study focuses on analyzing the shape feature distribution of rock particles and establishes a machine learning-based volume prediction model for rockfill materials. This approach characterizes the relationship between dimensional parameters and particle volumes to enhance gradation calculation accuracy.

2.1. 3D characteristics of rockfill particles

In practical engineering, rockfill dams typically utilize locally sourced materials processed on-site for construction. Due to variations in quarry sources and extraction techniques, rockfill particles exhibit highly diverse and irregular geometries [30,31]. To build high-precision volume prediction models, systematic and quantifiable characterization of particle morphology is essential. Current research classifies rock particle morphology into three scales: contour features defining overall geometric configuration that directly influences spatial occupancy; angularity features characterizing local edge sharpness that indirectly affects packing density and stability; and surface texture relating to microscopic roughness [32,33]. Given our focus on volume estimation—where surface texture has minimal impact—we extract the first two feature categories as model inputs.

(1) Contour features

Contour features reflect morphological properties along principal dimensions. These dimensions comprise three mutually perpendicular axes: the major axis (L), intermediate axis (W), and minor axis (T), corresponding to the length, width, and height of a particle's minimum bounding box. Principal Component Analysis (PCA) is commonly employed to compute these axes through orthogonal transformation that projects particle point clouds onto directions of maximum variance. This process aligns coordinate axes with principal dimensions, enabling determination of axial lengths from coordinate range differences along each axis, as illustrated in Fig. 1.

Elongation (El) and flatness (Fl) are two fundamental metrics for characterizing particle shape morphology. Elongation quantifies particle slenderness, defined as the ratio of the major axis to the intermediate axis ($El = L/W$), where increasing values indicate more elongated forms. Flatness describes particle platyness, calculated as the ratio of the minor axis to the intermediate axis ($Fl = T/W$), with higher values reflecting greater thickness. The computational expressions are formulated as follows:

$$El = \frac{L}{W} \quad (1)$$

$$Fl = \frac{T}{W} \quad (2)$$

Sphericity (Sp) quantifies the degree to which a rock particle resembles an ideal sphere, with values ranging between 0 and 1. A sphericity value of 1 indicates a perfect spherical shape, while decreasing values reflect greater deviation from sphericity. Mathematically, sphericity is defined as the ratio of the particle's surface area to the surface

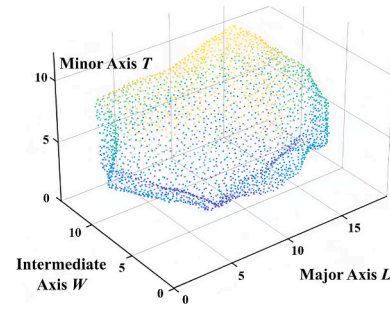


Fig. 1. Computation of principal particle dimensions.

area of a sphere with equivalent volume, expressed as:

$$Sp = \frac{\sqrt[3]{36\pi V^2}}{S} \quad (3)$$

Where V and S denote the volume and surface area of the rock particle, respectively.

To compute V and S from rock particle point clouds, this study employs either Delaunay triangulation or Poisson surface reconstruction to construct the particle's surface mesh model. Volume calculation adopts a tetrahedral decomposition method, which subdivides the mesh into tetrahedral units for volumetric summation. For a given triangular mesh, let the position vectors of any triangular face vertices be $\vec{v}_1, \vec{v}_2, \vec{v}_3$. Using the origin as the fourth vertex of the tetrahedron, the volume of a single tetrahedral unit is calculated as:

$$V_i = \frac{1}{6} \left| \vec{v}_1 \cdot (\vec{v}_2 \times \vec{v}_3) \right| \quad (4)$$

The total volume V of the entire mesh is computed as the summation of all tetrahedral volumes:

$$V = \sum_{i=1}^N V_i \quad (5)$$

The surface area S is obtained by accumulating the areas of all triangular facets:

$$S_i = \frac{1}{2} \left| (\vec{v}_2 - \vec{v}_1) \times (\vec{v}_3 - \vec{v}_1) \right| \quad (6)$$

$$S = \sum_{i=1}^N S_i$$

(2) Angularity feature.

Angularity features characterize the local contour properties of rock particles, specifically quantifying the sharpness or roundness of edges and corners [34]. In existing research, roundness (Rd) serves as a fundamental metric for assessing corner bluntness. Wadell et al. defined 3D roundness as the average ratio of the curvature radius at each corner to the radius of the maximum inscribed sphere, with values ranging between 0 and 1. Higher values indicate more rounded edges [35]. The computational formula is expressed as:

$$Rd = \frac{\sum_{i=1}^N r_i}{R} / N \quad (7)$$

In the formula: r_i is the radius of the sphere fitted to the i -th corner; R is the radius of the maximum-inscribed sphere of the particle; N is the number of corners of the particle.

To achieve the roundness calculation under three-dimensional point cloud data, this paper first uses the Delaunay triangulation algorithm to construct the surface mesh model of the particle and calculates each parameter in turn.

The radius R of the maximum-inscribed sphere of the particle is calculated by the Euclidean distance field method: First, uniformly

sample inside the particle to generate an internal point set P_{in} . For each internal sample point $p \in P_{in}$, find its minimum Euclidean distance R_p to the outer surface of the particle. Then, the maximum value of the minimum distances corresponding to all sample points is the radius R of the maximum-inscribed sphere.

$$\begin{aligned} R_p &= \min_{q \in S_{null}} \|p - q\| \\ R &= \max(R_p) \end{aligned} \quad (8)$$

In the formula: S_{null} is the surface-point set; q is any point on the surface.

Calculation of the fitted-sphere radius of particle corners: First, it is necessary to identify the corner regions based on curvature information, and then calculate the corresponding fitted-sphere radius [36]. The curvature of the particle surface is estimated by locally fitting a surface using the least-squares method. The fitted surface equation and the corresponding Hessian matrix are as follows:

$$f(u, v) = a_1x^2 + a_2y^2 + a_3xy + a_4x + a_5y + a_6 \quad (9)$$

$$H = \begin{bmatrix} 2a_1 & a_3 \\ a_3 & 2a_2 \end{bmatrix} \quad (10)$$

The principal curvatures k_1 and k_2 are calculated from the eigenvalues of the Hessian matrix, and the mean curvature is then:

$$k_m = \frac{k_1 + k_2}{2} = \frac{a_1 + a_2}{2} \quad (11)$$

By comparing the curvature of all points with the curvature of the maximum-inscribed sphere, the angular regions can be determined:

$$k_i > \frac{1}{R} \quad (12)$$

After determining the angular regions, calculate the radius of the fitting sphere for the points within this region. Finally, according to eq. (7), the final particle roundness index can be calculated.

In addition, the form factor (Ff) is also an important index for describing the angular features of the two-dimensional contour of particles. It is defined as the ratio of the circumference of an equivalent circle with the same projected area as the particle to the projected perimeter, and the calculation formula is as follows:

$$Ff = 4\pi A / P^2 \quad (13)$$

In the formula: A is the two-dimensional contour area of the particle, that is, the area of the equivalent circle; P is the two-dimensional contour perimeter of the particle.

2.2. Rock volume prediction based on 3D features and LightGBM

Following feature extraction, the next step establishes mapping relationships between these features and actual volumes. Traditional image-based methods predominantly estimate volume from 2D projections, neglecting height dimensions and introducing systematic errors. In contrast, 3D point clouds provide more comprehensive geometric information, enabling higher-precision volume prediction models.

Given the moderate-scale dataset (400 samples) and structured numerical features in this study, traditional machine learning models outperform deep neural networks in training efficiency, interpretability, and generalization capability. Therefore, we employ the LightGBM (Light Gradient Boosting Machine) framework to construct and train the particle volume prediction model.

LightGBM is an efficient ensemble-learning framework based on Gradient Boosting Decision Tree (GBDT). Its core idea is to iteratively train multiple weak learners (decision trees) to gradually fit the residuals and finally combine them into a strong prediction model [37]. A decision tree is a supervised-learning algorithm for making decisions

based on tree structures. It recursively divides the sample feature space through conditional judgments to achieve classification or regression prediction. GBDT is an ensemble-learning method based on the Boosting framework, and its core idea is to learn the residuals through multiple weak decision trees [38]. As shown in Fig. 2, in the GBDT algorithm, first, an initial decision-tree model $F_0(x)$ is generated to predict the target value. Then, the residual between the true value of each sample and the predicted value of the current model is calculated. Next, a new decision tree is trained to learn this residual, and it is multiplied by the learning rate η and added to the current model. This process is repeated continuously, adding new decision trees until the preset number of trees is reached or the error converges, so as to obtain the final prediction model.

Compared with the GBDT algorithm, LightGBM has made multiple optimizations in its structural design. On the one hand, LightGBM introduces the histogram algorithm [39]. Traditional decision trees find the optimal split point by traversing all data one by one, while LightGBM discretizes continuous feature values into k intervals and constructs a histogram. It only needs to traverse the histogram to find the optimal split point, which significantly reduces the computational complexity and improves the training speed. On the other hand, LightGBM adopts the Leaf-wise growth strategy [40]. Traditional decision trees expand all nodes at each level simultaneously, while LightGBM preferentially expands the leaf node with the largest gain (as shown in Fig. 3), which makes the loss decrease faster. With the same number of leaf nodes, the generated tree has a lower error. This model is especially suitable for processing small-and medium-sized structured data. It can maintain high performance and has good tunability, making it very suitable for the application scenario of this study.

2.3. Rock volume prediction based on 3D features and LightGBM

Although LightGBM demonstrates strong predictive capabilities, its black-box nature may undermine engineering confidence. To enhance model transparency, this paper employs SHAP (Shapley Additive Explanations) for interpretability analysis [41,42]. The core concept originates from the Shapley value in cooperative game theory, treating model predictions as collective outcomes of feature interactions while quantifying each feature's contribution through Shapley values [43,44]. These values enable interpretation of individual predictions by quantifying per-feature contributions, reveal global and local feature importance, and increase decision transparency by clarifying feature dependencies. Specifically, SHAP values reflect a feature's expected marginal effect across all possible feature combinations, mathematically expressed as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (14)$$

In the formula: N is the complete set of all features; S is any subset of N that excludes feature i .

3. Results

3.1. Analysis of 3D characteristics

To establish a reliable volume prediction model, this study prepared a particle point cloud dataset comprising 400 paired samples. Full 3D point clouds of rock particles were obtained through terrestrial laser scanning (TLS), with geometric reconstruction performed in Geomagic Wrap to generate watertight meshes for ground-truth volume calculation. Subsequently, single-view projection point clouds were extracted by orthographically projecting the complete 3D point clouds onto a horizontal plane from a top-down view, simulating the typical scanning geometry in field applications where only the upper surface of particles is visible due to occlusion. This process effectively mimics the "2.5D"

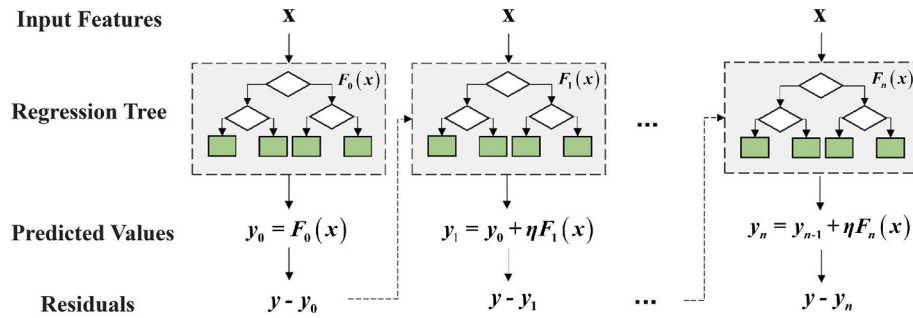


Fig. 2. Principle of GBDT algorithm.

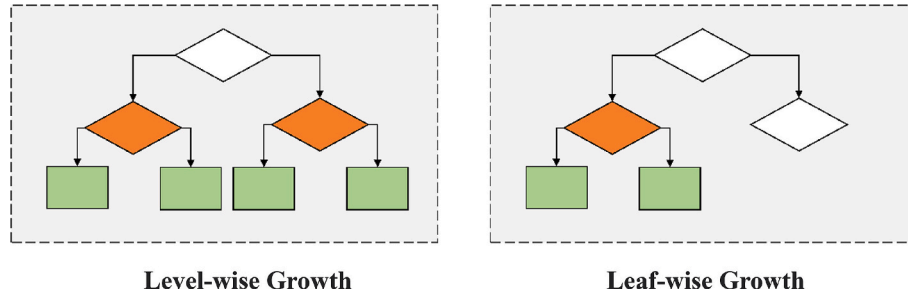


Fig. 3. Leaf-wise growth strategy.

nature of real-world laser scans and ensures that the incomplete morphology aligns with practical measurement conditions, forming matched 2.5D-3D particle pairs. Following noise filtering via statistical outlier removal and surface completion through Poisson reconstruction, morphological feature extraction generated a high-quality dataset containing volume (V), surface area (S), principal dimensions (L, W, T), shape indices (Sp, Rd), and angularity metrics (Ff). Key statistical descriptors for these parameters are documented in Table 1.

To quantify the geometric deviations caused by incomplete point clouds, this paper selects typical samples and compares their key volume and size parameters under 2.5D point clouds and 3D complete point clouds (Fig. 4 and Table 2). The results show that due to the lack of the lower-half part of the particles in 2.5D point clouds, the estimated volume is much lower than the real volume. The differences are particularly significant in the minor axis length and surface area, which further affects the accuracy of the overall volume calculation. This difference highlights the limitations of 2.5D point clouds in particle-size distribution analysis. Therefore, it is urgent to use modeling methods to compensate for the missing information and achieve the goal of predicting the complete particle volume from incomplete point clouds.

Table 1
List of shape parameters.

Serial Number	3D Features	2D Contour Features
1	Particle volume V	Projection area A
2	Particle surface-area S	Projection perimeter P
3	Major-axis of the minimum bounding box L	Length of the minimum-bounding rectangle l
4	Minor-axis of the minimum bounding box W	Width of the minimum-bounding rectangle w
5	Short-axis of the minimum bounding box T	Equivalent circle diameter d
6	Maximum inscribed-sphere radius R_1	Major-axis of the equivalent ellipse a
7	Minimum circumscribed-sphere radius R_2	Minor-axis of the equivalent ellipse b
8	Sphericity Sp	Sphericity Sp_1
9	Roundness Rd	Form Factor Ff

In the constructed dataset, a statistical analysis was carried out on the main geometric features such as the elongation (El), flatness (Ff), sphericity (Sp), and roundness (Rd) of 400 groups of particles. As shown in Fig. 5, the elongation of the 2.5D particles of the stockpiled materials is widely distributed in the range of 1–2.2, the flatness is between 0.2–1, the sphericity is concentrated in the range of 0.5–0.9, and the roundness is mainly distributed in the range of 0.2–0.6.

Further summarize the mean values of shape characteristics within different particle size intervals (Table 3). The results show that each parameter does not change significantly within different particle size ranges, and the shape distribution shows good consistency. This indicates that the particles of the same batch of rockfill materials have a highly consistent morphological characteristic distribution, which lays a solid foundation for establishing a general prediction model across particle sizes.

3.2. Performance evaluation of volume prediction model based on three-dimensional features

When using LightGBM for regression task training, multiple hyperparameters need to be set, and the values of these hyperparameters often affect the prediction performance of the model. To select the best hyperparameters for optimal model prediction, this paper uses the Optuna algorithm for hyperparameter optimization. Optuna is a hyperparameter tuning framework based on improved Bayesian optimization [45]. It can dynamically define the hyperparameter search space, estimate the distribution of each hyperparameter by constructing a probability model, and thus efficiently perform search and pruning operations. It can predict the next optimal sampling point based on the current performance of the model and terminate poorly-performing models in advance, thereby accelerating the hyperparameter optimization process. The specific hyperparameters and their optimization results are shown in Table 4.

To verify the performance of the particle volume prediction model based on three-dimensional features, this paper selects four common error evaluation indicators: the correlation coefficient (R^2), the mean absolute error (MAE), the root mean square error ($RMSE$), and the mean absolute percentage error ($MAPE$) to evaluate the model's performance

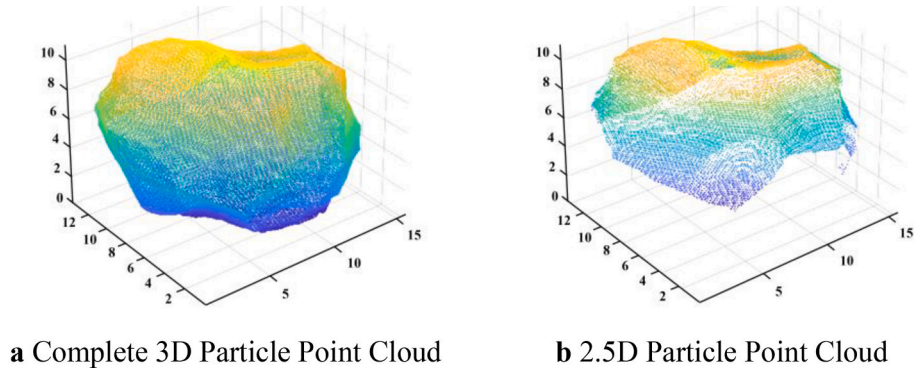


Fig. 4. Complete 3D particles and 2.5D particle point cloud shapes.

Table 2

Comparison of three-dimensional sizes between 3d particles and 2.5D particles.

Three-dimensional Feature	3D Particle Point Cloud	2.5D Particle Point Cloud
major axis $L(\text{mm})$	15.7	15.1
intermediate axis $W(\text{mm})$	12.8	12.6
minor axis $T(\text{mm})$	11.1	7.4
Surface area $S(\text{mm}^2)$	537.1	429.9
Volume $V(\text{mm}^3)$	1035.3	607.8

from different perspectives [46]. R^2 reflects the correlation between the predicted values and the true values, and the remaining three measure the error level. The smaller the value, the better the performance. The calculation formulas of each evaluation index are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (15)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (16)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (17)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\% \quad (18)$$

Among them, y_i represents the i -th true value in the test set, $\hat{y}_i$ represents the i -th predicted value of the model, and $\bar{y}$ represents the mean of the true values in the test set.

A volume prediction model was constructed using the nine two-dimensional contour features listed in Table 1 — including projection area (A) and projection perimeter (P) — with hyperparameters optimized by Optuna, and subsequently trained. Following training, the model was evaluated on an independent test set of 50 sample particles. The comparison between predicted and true particle volumes is presented in Fig. 6. To assess the effectiveness of different feature sets in predicting complete particle volumes, four evaluation metrics were applied to compare the performance of models built upon three-dimensional geometric features versus those based on two-dimensional contour features, as summarized in Table 5.

The analysis results show that the three-dimensional features have a stronger correlation with the particle volume. In contrast, due to the lack of particle height information, it is more difficult for two-dimensional contour features to accurately predict the three-dimensional volume of particles. Especially for particles with larger particle sizes, more height information will be lost, resulting in larger prediction errors. The three-dimensional features contain complete spatial scale information, which effectively alleviates the uncertainty caused by irregular shapes and

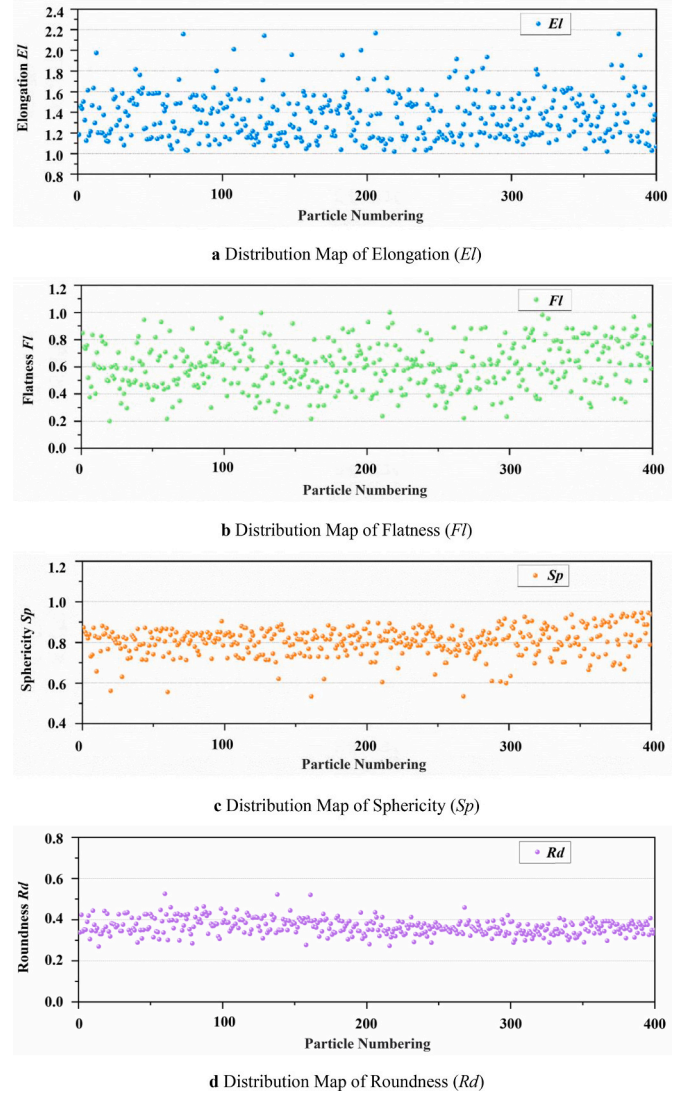


Fig. 5. Distribution diagram of particle shape characteristics of rock-fill materials.

significantly improves the prediction robustness.

3.3. Model feature analysis

To further reveal the internal mechanism of the model, this paper uses the SHAP method to analyze the feature importance of the three-

Table 3
Statistics of 2.5D particle shape parameters of rockfill materials.

Particle-size Interval (mm)	Mean Elongation El	Mean Flatness Fl	Mean Sphericity Sp	Mean Roundness Rd
5–20	1.371	0.583	0.796	0.374
20–40	1.345	0.590	0.799	0.387
40–60	1.357	0.594	0.802	0.361
60–80	1.354	0.573	0.792	0.353
80–100	1.375	0.669	0.828	0.355

Table 4
Results of optuna hyperparameter optimization.

Hyperparameter	Range	Result
learning_rate	[0.01,0.5]	0.1
max_depth	[−1,16]	6
max_bin	[25,255]	89
num_leaves	[2,50]	36
min_child_samples	[1,50]	4

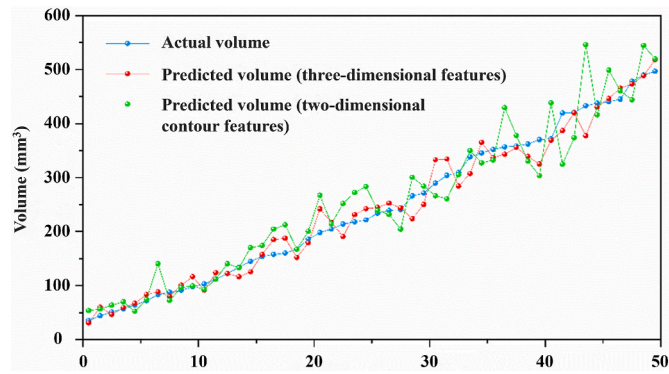


Fig. 6. Volume prediction results based on different features.

Table 5
Calculation results of evaluation indicators.

Evaluation Indicators	R^2	MAE	RMSE	MAPE
Three-dimensional Features	0.979	0.459	0.724	9.39%
Two-dimensional Features	0.925	0.844	1.389	15.19%

dimensional feature parameters for the particle volume V . Fig. 7 shows the ranking of the contribution of each feature to the model prediction results, that is, the global feature importance. It can be seen that the 2.5D particle volume V and the particle surface area S are the two most critical variables affecting the prediction results, and their SHAP values are significantly higher than other geometric parameters. Subsequently, they are followed by the short axis T of the minimum bounding box, the radius R_1 of the largest inscribed sphere, etc.

To further explore the specific relationships between each feature and the predicted value, Fig. 8 shows a feature summary plot based on SHAP values. In this plot, each row on the vertical axis represents a feature variable, and each point corresponds to a sample. The color of the point reflects the magnitude of the feature value, while the horizontal axis represents the distribution of the feature's influence on the model output. A positive SHAP value indicates a positive influence, and a negative value indicates a negative influence. As can be seen from the figure, the nine three-dimensional feature parameters generally show a distribution with blue on the left and red on the right, indicating that larger feature values have a positive influence on the prediction results, increasing the volume prediction value, while lower feature values reduce the volume prediction value. Among them, the volume and

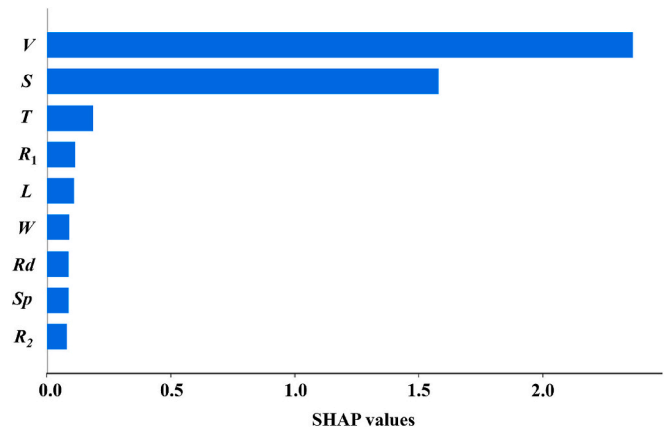


Fig. 7. Feature importance analysis.

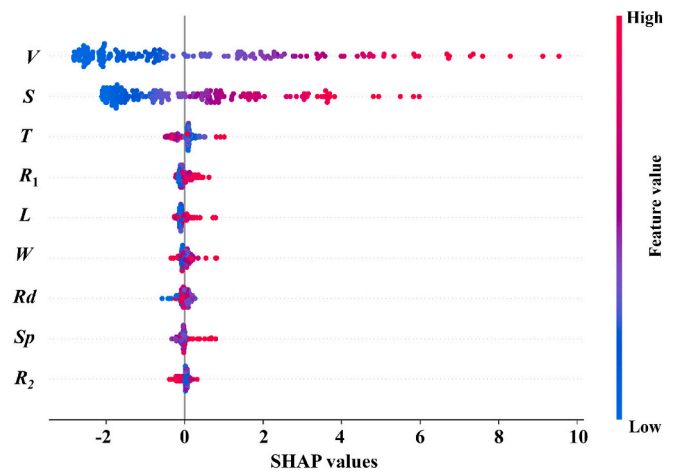


Fig. 8. Feature summary diagram.

surface area of 2.5D particles are significantly positively correlated with the predicted value of the actual volume, indicating that the high-dimensional features in the three-dimensional features have an excellent effect on volume prediction, significantly improving the accuracy of volume prediction.

3.4. Application verification of the volume prediction model for gradation detection

To verify the practical application value of the proposed method, this paper embeds the volume prediction model into the complete gradation detection process: First, the 2.5D point cloud of a single particle is extracted through point cloud instance segmentation. It should be noted that, in real-world scenarios, instance-level segmentation of densely packed rock particles remains challenging due to occlusion and contact-induced merging. To isolate the performance of the volume prediction model from segmentation errors, we adopted synthetically generated stockpiles with ground-truth segmentation masks derived directly from depth map labeling. This approach ensures pixel-perfect separation of adjacent particles while preserving realistic occlusion patterns, thereby enabling a clean evaluation of the downstream volume estimation and gradation analysis pipeline. Then, its three-dimensional geometric features are calculated and input into the trained volume prediction model to estimate its complete volume. Next, the particles are classified by particle size according to the long axis L , and the volume proportion of each particle size interval is counted. Finally, the gradation curve is generated.

It is worth noting that in real working conditions, the point cloud segmentation process may introduce boundary misjudgment due to particle adhesion or occlusion, which in turn affects the accuracy of subsequent volume prediction and gradation calculation. To eliminate such interference and independently evaluate the performance of the volume prediction model itself, this paper conducts experimental verification using the point cloud data obtained from the mapping of the depth map labels of the virtually generated rockfill materials. The virtual rockfill materials were generated by simulating realistic particle morphologies based on statistical distributions derived from the measured 2.5D–3D paired dataset. Specifically, particle sizes, shapes, and spatial arrangements were sampled to match the observed characteristics of real rockfill. Using these parameters, synthetic 3D particles were modeled with consistent geometric fidelity, and their top-view projections were rendered into depth maps. These depth maps were then converted into 2.5D point clouds via inverse projection, preserving occlusion patterns typical of real-world scanning scenarios while guaranteeing accurate ground-truth segmentation labels. At the same time, the long axis L is uniformly used as the standard for the particle size D in the gradation calculation to avoid additional deviations caused by different definitions of particle size. Fig. 9 shows the specific process of mapping the depth map of rockfill materials to the point cloud.

To validate the robustness of the proposed volume prediction model under diverse gradation scenarios, six distinct synthetic stockpile datasets were independently generated using realistic morphological distributions derived from the measured 2.5D–3D paired particle dataset. Each group was constructed by randomly sampling particle sizes, shapes, and spatial arrangements according to statistically representative parameters stratified across five size intervals, ensuring varied yet physically plausible gradation profiles. Based on the segmentation results, the three-dimensional geometric features of 2.5D particles were extracted and input into the trained volume prediction model to estimate their complete volumes. On this basis, the cumulative volume distribution was statistically analyzed according to the particle size intervals, and six groups of gradation detection results were obtained, as shown in Table 6.

As shown in Table 6, the detection errors of the six groups of samples in each particle size range remained at a low level. The maximum errors were 0.92%, 1.86%, 2.55%, 2.84%, and 4.74% respectively, and the average absolute errors were 0.31%, 0.77%, 1.91%, 1.79%, and 1.41% respectively. The overall average error was controlled within 2%, indicating that the gradation data obtained based on three-dimensional feature detection was highly consistent with the true values. As shown in Fig. 10, the six groups of gradation curves almost coincided with the

true curves, indicating that this method has excellent reproducibility.

4. Conclusions

This paper focuses on the problem of estimating the complete volume of 2.5D rock particles and proposes a new method for detecting the gradation of rockfill materials by integrating interpretable machine learning. The research shows that:

- (1) Through the statistical analysis of 400 groups of 2.5D particles, it is found that the rockfill materials in the same batch have a high degree of consistency in shape characteristics such as acicularity, flatness, sphericity, and roundness, which provides a prerequisite for establishing a general prediction model across particle sizes.
- (2) Compared with two-dimensional contour features, the three-dimensional geometric features (especially volume and surface area) extracted from 2.5D point clouds can more effectively characterize the spatial morphology of particles. The constructed LightGBM model achieves an excellent performance of $R^2 = 0.979$ on the test set, significantly reducing the systematic deviation caused by information loss.
- (3) Validation on six groups of synthetically generated rockfill datasets, designed to mimic real morphological characteristics, demonstrated that detection errors across all gradation intervals remained below 2%. This confirms the model's potential for high-accuracy application, though future work will focus on validating the framework under real-field scanning conditions. The gradation curve is in good agreement with the real gradation curve, meeting the engineering accuracy requirements.

CRediT authorship contribution statement

Jiayu Wang: Writing – review & editing, Writing – original draft, Conceptualization. **Mengnan Shi:** Writing – review & editing, Supervision, Methodology. **Yilong Wu:** Investigation, Funding acquisition. **Peng Ji:** Writing – review & editing, Resources. **Hongtao Li:** Writing – review & editing, Supervision. **Qiang Yao:** Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

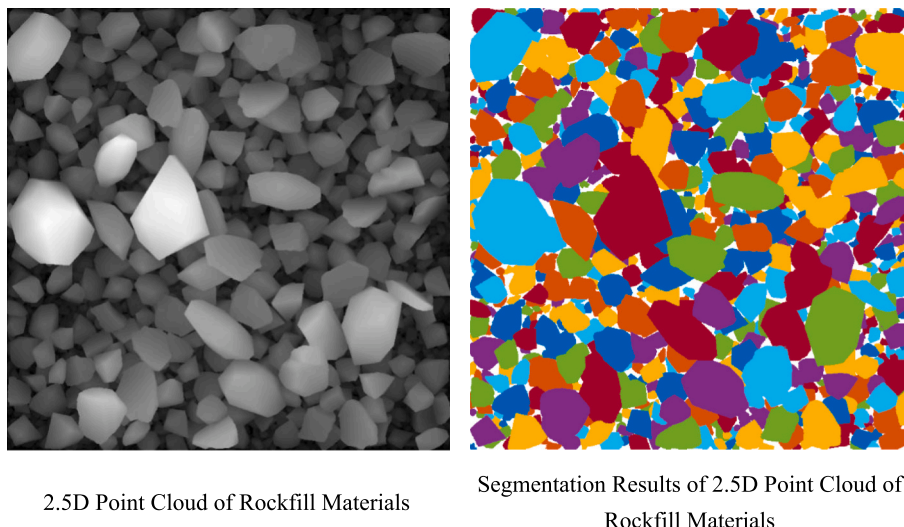


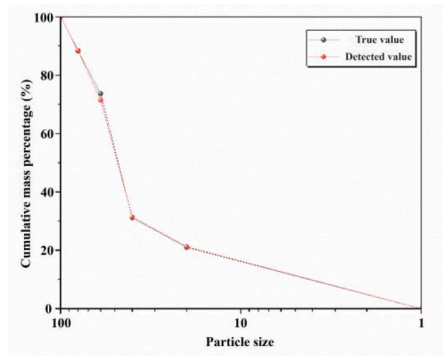
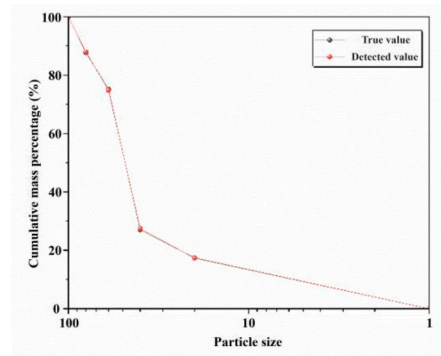
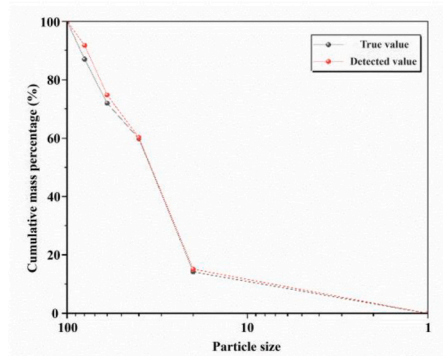
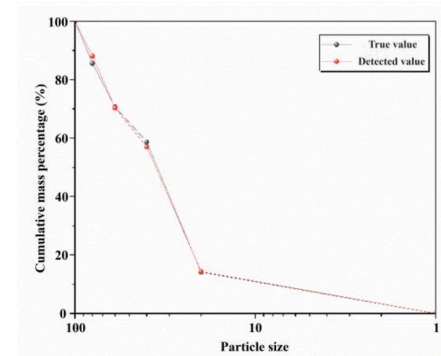
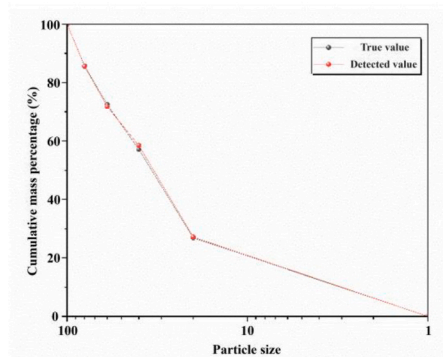
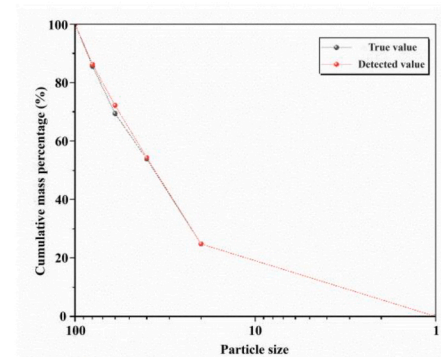
Fig. 9. Depth map-mapped point cloud of rockfill materials.

Table 6

Gradation detection results of rockfill materials based on three-dimensional features.

SN	1–20 mm		20–40 mm		40–60 mm		60–80 mm		80–100 mm	
	TV	DV	TV	DV	TV	DV	TV	DV	TV	DV
1	21.01%	21.26%	10.12%	10.19%	42.55%	40.00%	14.55%	16.98%	11.77%	11.57%
2	17.46%	17.32%	9.58%	10.14%	48.17%	47.33%	12.42%	13.16%	12.37%	12.05%
3	14.23%	15.15%	45.56%	45.02%	12.20%	14.59%	15.02%	16.99%	13.00%	8.26%
4	14.12%	14.37%	44.55%	42.68%	12.01%	13.27%	14.87%	17.71%	14.45%	11.96%
5	26.90%	27.18%	30.21%	31.31%	15.37%	13.27%	13.16%	13.78%	14.36%	14.45%
6	24.79%	24.77%	29.08%	29.55%	15.55%	17.86%	16.17%	14.02%	14.41%	13.79%

Note: SN: Serial Number; TV = True Value; DV = Detected Value.

**a** Gradation Curve of the First Group of Rockfill Materials**b** Gradation Curve of the Second Group of Rockfill Materials**c** Gradation Curve of the Third Group of Rockfill Materials**d** Gradation Curve of the Fourth Group of Rockfill Materials**e** Gradation Curve of the Fifth Group of Rockfill Materials**f** Gradation Curve of the Sixth Group of Rockfill Materials**Fig. 10.** Gradation test results of rockfill materials.

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Data availability

Data will be made available on request.

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